

Ecological interface design effectively reduces cognitive workload – The example of HMIs for speed control

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ABSTRACT

This paper aims to demonstrate that ecological interface design reduces the use of cognitive resources. Making the boundaries of acceptable performance visually perceivable should elicit skill-based behavior, thus lowering the cognitive load. To illustrate the psychological mechanisms of ecological interface design and validate its influence on cognitive load, we compared a conventional speedometer to an ecological speedometer. Both interfaces were displayed on the head-up display of a static driving simulator and were tested in a rural highway scenario. In a $2 \times 2 \times 2$ repeated-measure design, the human-machine interface was tested along with the informative intelligent speed assistance system (representing an existing alternative for speed control) and the additional workload induced by a 1-back task (representing cognitively demanding driving situations). To measure cognitive load, we used a tactile version of the detection response task and controlled for all confounding variables. The experiment was conducted with 28 male and 21 female drivers. Our hypotheses and the ecological interface design theory are supported by the finding that the ecological speedometer reduces the use of cognitive resources. Moreover, the specific application of the ecological speedometer enables better driving performance (i.e., longitudinal and lateral control) compared to a conventional speedometer.

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1. Introduction

Advances in hardware and software provide engineers with the possibility to design interfaces that can provide drivers with a substantial amount of information and entertainment. However, increased input might also increase the driver's cognitive load, which is defined as the demand for cognitive control (Engström et al., 2013; International Organization for Standardization [ISO], 2016). From a more neuroscientific perspective, automatized behavior relies on strong neural pathways trained on perceptual-motor contingencies with valuable and consistent outcomes, whereas nonautomatized behavior builds on weaker pathways caused by varied and relatively infrequent contingencies (Cohen, Dunbar, & McClelland, 1990; Miller & Cohen, 2001; Shiffrin & Schneider, 1977). Following the cognitive control hypothesis proposed by Engström, Markkula, Victor, and Merat (2017), cognitive control is needed to flexibly boost these weaker neural pathways to allow for nonautomatized and goal-directed behavior. Therefore, cognitive resources that are demanded by any nonautomatized secondary task specifically impair tasks that are also nonautomatized and thus share the demand for cognitive control. However, because cognitive control is not needed, automatized performance is unaffected by the induced cognitive load.

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Whenever the cognitive load becomes too high, weaker pathways can no longer be flexibly boosted, and the stronger pathways serve as a kind of fallback level. For example, cognitively loaded drivers rely on this fallback level and would most likely brake sufficiently before an obstacle because looming effects automatically trigger a braking reaction (Franconeri & Simons, 2003). However, cognitively loaded drivers might not be able to properly adapt to predictive cues, such as road signs and roadside instructions (Baumann, Petzoldt, Hogema, & Krems, 2008; Engström & Markkula, 2007) or dangerous road conditions (Cooper et al., 2003). Moreover, capture errors, as described by Norman (1981), might happen more often to cognitively loaded drivers, when a partially familiar situation unconsciously activates an inappropriate automatized behavior. Thus, if the automatized behavioral patterns are sufficient to handle a given situation, cognitive load generates no safety hazard. However, whenever the automatized behavior cannot handle a given situation properly, severe accidents might occur (Engström et al., 2017).

Therefore, to save on scarce cognitive resources and prohibit possibly dangerous automatized behavior, the literature suggests ecological interface design as an effective framework for designing interfaces in a way that theoretically minimizes the cognitive load (Vicente, 2002). More specifically, ecological interface design is an approach that seeks to display information about the constraints and complex relationships of the work environment in a manner that enables information to be directly perceived and easily understood (Hajdukiewicz & Burns, 2004).

A good representation of a system simultaneously visualizes the relevant functions, goal states, current states, functional consequences of time, and critical boundaries for action (Vicente, 1999). In Rasmussen (1983) framework of skills, rules, and knowledge (SRK), ecological interfaces enable a direct shortcut from “activation” to “execution” because they provide consistent perception–action mappings and thus can be used as signals (Fig. 1). This indicates that the information shown by the ecological interface should lead to the automated execution of a certain behavior, without having to first identify and interpret the information and then decide on the appropriate task in the particular situation. Thus, ecological interfaces are aimed at fostering skill-based behavior, which is the most automated and least conscious type of behavior. This theoretically reduces cognitive load, which then can be used to handle other tasks that demand higher cognitive processes.

Following Rasmussen’s model, sensory input has the characteristics of either signals, signs, or symbols. Signals simply represent time–space variables of a spatial and dynamic environment. They allow for direct perception–action coupling, which is unconscious and generates no cognitive load. In contrast, signs are features of the environment that refer to learned rules. Signs cannot be processed subconsciously because they are meant to activate rule-based behavior that can be verbal-

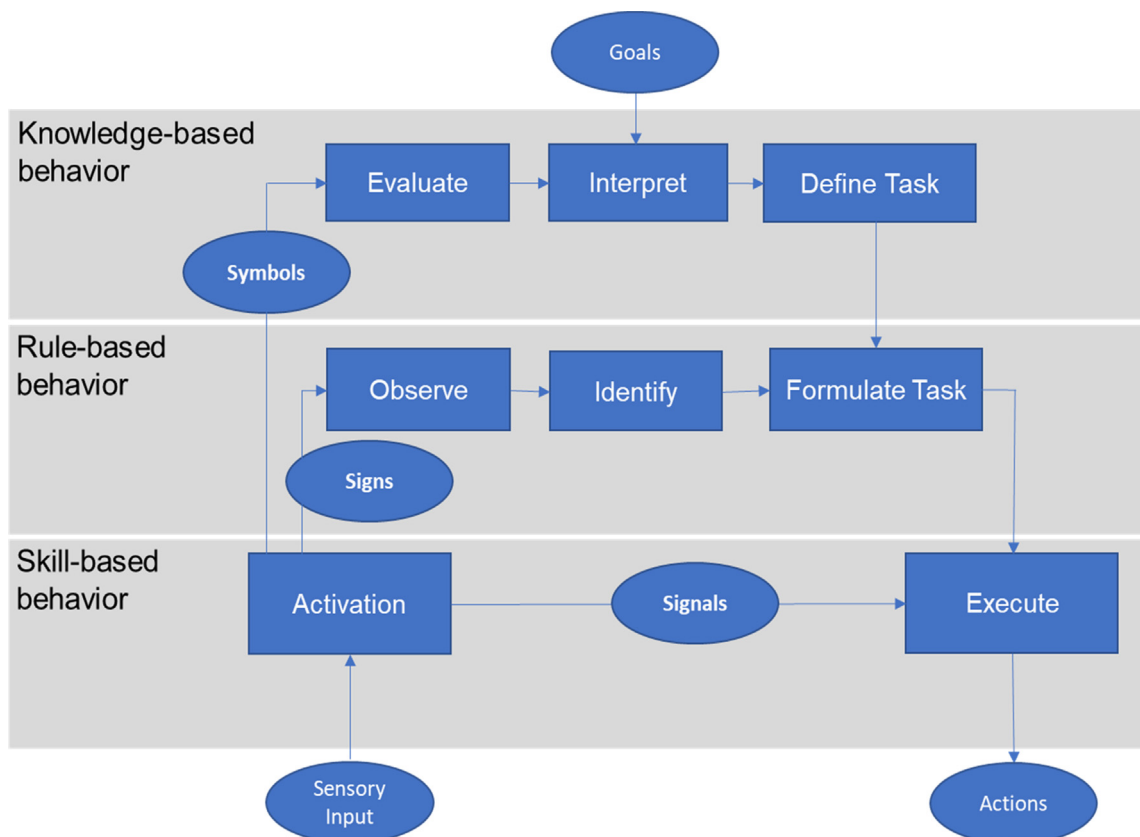


Fig. 1. Skills, rules, and knowledge (SRK) taxonomy. Adapted from Rasmussen, Pejtersen, and Schmidt (1990).

ized. Lastly, symbols represent further information that must be formally processed to evaluate an existing problem in a given situation. This kind of behavior is the most conscious behavior and called knowledge-based behavior.

Commonly, an abstraction hierarchy is used to determine which kinds of information should be displayed on a system's ecological interface (e.g., [Jenkins, Stanton, & Walker, 2017](#)). Then, knowing which information is relevant in a certain domain, how it should be presented is determined using the SRK taxonomy from Rasmussen. For example, as discussed by [Borst, Suijkerbuijk, Mulder, and Van Paassen \(2006\)](#), an ecological interface should reveal the affordances of the work domain in a way that supports all three levels of cognitive control. Therefore, the interface should provide signals, signs, and symbols.

To demonstrate the approach of the ecological interface design, we decided to use the example of a human-machine interface (HMI) for speed control. Because speed control is vital for road safety ([Elvik, Christensen, & Amundsen, 2004](#); [Kloeden, Ponte, & McLean, 2001](#); [Kröyer, 2015](#)) and advances in infotainment and in-vehicle information systems have increased the cognitive load of drivers ([Engström, Johansson, & Östlund, 2005](#)), conventional speedometers may need to be replaced by ecological speedometers. Regarding speed control, no work domain analysis needs to be conducted because it is a basic task that can be executed on different cognitive control levels, either automated or nonautomated. For speed control, both control and set values should be presented.

We consider speed control that occurs while driving a vehicle on a rural highway. While maintaining a certain speed relies on signals, such as optical flow, the driver must use the speedometer to adapt to new speed limits. The usual speedometer presents the current speed as a number that must be cognitively compared to the speed presented on road signs or the HMI of the vehicle. If the current speed is higher, the driver must reduce speed. If it is lower, he or she may increase the speed. In this case, the speedometer is used as a sign, causing nonautomated behavior. Theoretically, such a conventional speedometer cannot be used as a signal because it does not spatially and consistently represent the necessary changes in speed to adjust to a certain limit. Although maintaining a certain speed (stabilization) becomes skill-based after some training, the artificial adaption to certain speed limits (guidance) remains rule based ([Donges, 1999](#)). This idea is also in line with the cognitive control hypothesis ([Engström et al., 2017](#)), where cognitive control is needed to use variable mapped cues, such as speed signs, whereas consistently mapped perceptual-motor contingencies enable automated behavior. For example, maintaining lanes is based on optical flow and the perceivable boundaries of the street and thus seen to be automatized. As [Engström et al. \(2017\)](#) suggested, even general stimulus-response contingencies may allow for the development of specific automaticity. For example, the ability to unconsciously keep a certain distance to a preceding car (car-following based on optical parameters), may be based on general contingencies learned from walking. Regarding speed control, [Recarte and Nunes \(2002\)](#) argued that every driver's specific set of automatized behaviors determines a certain speed for a given task demand. However, if no other consistent stimuli exist (e.g., given by an ecological interface), conscious deviations from this individual optimum still rely on cognitive control and contribute to the cognitive load. Under cognitive load, drivers likely return to their automatized behavioral patterns. They devote a major share of cognitive control to the secondary task while unconsciously following their automatized optimum speed, thus ignoring speed signs. This mechanism might also be the reason for unintentional speeding ([Lewis-Evans, de Waard, & Brookhuis, 2011](#); [Recarte & Nunes, 2002](#)).

To design an interface that provides consistent stimulus-response contingencies and therefore can be safely used in a skill-based automated manner, [Schewe and Vollrath \(2019\)](#) used the SRK taxonomy to develop an ecological interface for speed control. A simulated head-up display was used to augment a blue carpet and a bracket in front of the vehicle. The current speed was used to adjust the length of the carpet. The appropriate speed (speed limit) was shown as the augmented

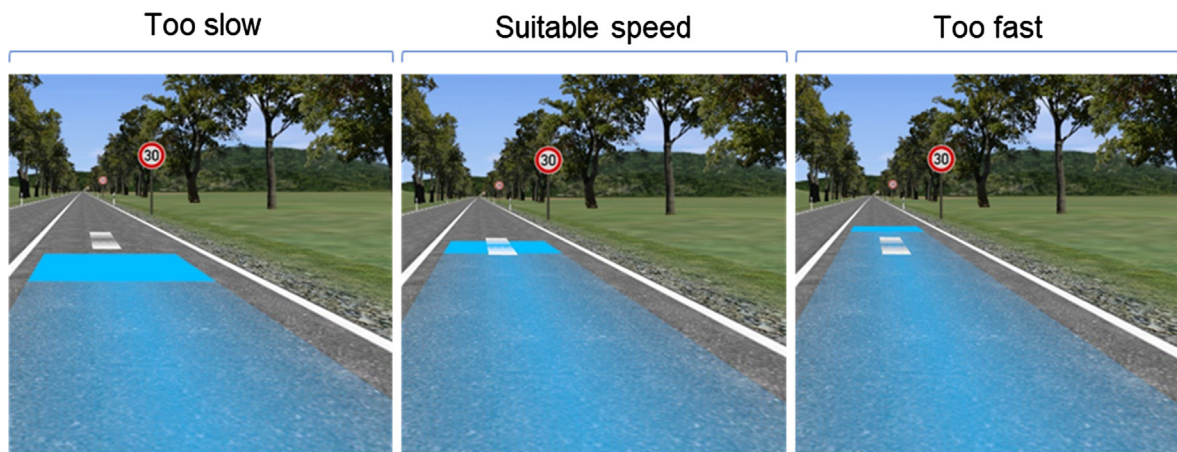


Fig. 2. Different states of the ecological interface; adapted from [Schewe and Vollrath \(2019\)](#). The current speed is visualized as a semi-transparent carpet that ends in a thick horizontal bar (originally blue), and this bar should be within the bracket (originally white). The figure's contrast was enhanced for better readability in gray-scale printouts. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

bracket. As Fig. 2 illustrates, the carpet ended before the bracket if the driver was driving too slowly (left). It extended over the bracket if the driver was driving too fast (right), and it matched the bracket when the speed was suitable (middle).

Fig. 3 explains how the conventional and ecological distance speedometer communicate on the three different levels of the SRK taxonomy (Rasmussen, 1983). Both HMIs present information as symbols and signs, but the ecological interface can also communicate in the form of signals because it spatially visualizes the speed changes required by the driver. Preexisting stimulus–response contingencies, learned through car-following and other driving situations, are likely to support the automatized processing of the distance speedometer's signals.

Schewe and Vollrath (2019) demonstrated that the ecological distance speedometer is well suited for speed control. When reacting to speed signs, the longitudinal driving performance was comparably precise for both the conventional and ecological interface. However, the visual fixation time on the conventional speedometer was higher in comparison to the fixation time on the ecological distance speedometer. Moreover, when reacting to the speed changes of a vehicle ahead, the variance of the speed deviations was significantly lower with the help of the ecological distance speedometer. Nonetheless, this study was unable to determine whether the ecological interface was effective because it enabled skill-based behavior or simply because it displayed the relevant information better by continuously providing direct information about the appropriate speed. If the ecological interface enables skill-based behavior, the speed control should happen unconsciously and thus generate less cognitive load compared to more conscious rule-based behavior generated by the conventional speedometer.

To compare the cognitive load generated by the conventional speedometer or the ecological speedometer, both interfaces were shown in the head-up display. To experimentally vary the requirements of the driving situations, additional cognitive load was artificially induced using the 1-back task (Mehler, Reimer, Coughlin, & Dusek, 2009). Additionally, to allow for a comparison with an existing speeding countermeasure, an informative intelligent speed assistance (ISA) was tested as well. When activated, it issued a warning if the rate of speed was 5 km/h or more too fast. According to Brookhuis and de Waard (1999), the perceived level of the cognitive load does not change when using an ISA system. However, theoretically, such a warning still requires the skill-based control of one's own speed with the help of a conventional speedometer. It just reminds the driver to devote cognitive control to his or her speed control. In summary, the presented setup was designed to test

		
Symbol (Information that must be formally processed) If braking does not cause any effect to the visualization on the speedometer → speculate if the speedometer is defect or the brake does not function (functional thinking).	✓	✓
Sign (Features in the environment with reference to certain conventions for acts) If the visualized set speed is higher than the speed visualized on the speedometer → decelerate.	✓	✓
Signal (Sensory data which directly represents time-space variables from the dynamic spatial configuration in the environment. Boundaries of acceptable performance are directly perceivable) Keep the set point → Use the deviation from this set point as an error signal and track continuously.	✗	✓

Fig. 3. Information quality. Comparison of the two human–machine interfaces regarding elements acting as symbols, signs, or signals. Structure adapted from Rasmussen (1983).

whether the ecological interface reduces the cognitive load in situations with and without an ISA system and in situations in which additional cognitive load was induced.

To measure the attentional effects of cognitive load, a tactile version of the detection response task (DRT) was used, as described in the [ISO Standard 17488 \(2016\)](#). The task is an indirect measure of cognitive load in the sense that better performance in the DRT indicates that more resources could be used for this task, which indicates that the primary task generates less cognitive load and control demand. This follows the idea of limited cognitive resources according to [Wickens' resource theory \(2002\)](#) combined with the theory that a specific resource for cognitive control is necessary to handle nonautomatized tasks ([Nilsson, Aust, Engström, Svanberg, & Lindén, 2018](#)).

In the following paragraphs, we summarize the conceptual framework for understanding driver attention and cognitive load, as presented in the [ISO Standard 17488 \(2016\)](#). Driver attention can be conceptualized as the bottom-up and top-down allocation of resources to certain activities. These resources, which are allocated to certain activities, can be divided into the following three tiers: sensory/actuator resources, perceptual/motor resources, and cognitive control (on top). Transferred to the comparison between a conventional speedometer and an ecological distance speedometer, the allocation of resources should theoretically appear as they are depicted in [Fig. 4](#).

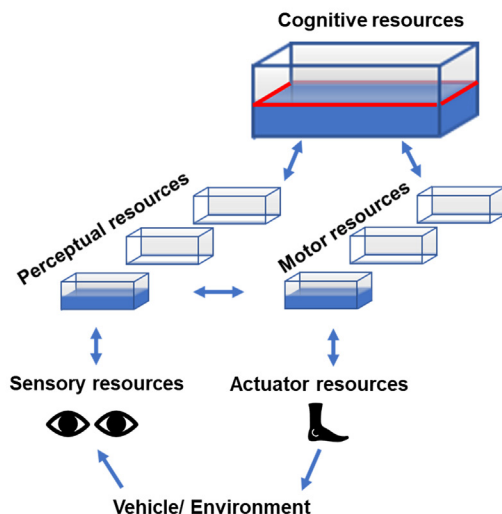
The figure visualizes resources as boxes that are filled according to the demand for the resource. The differences might not be large because speed control is an easy task, but the effects should nevertheless be detectable and stable, if the theory is correct. Both perceptual and sensory resources and motor and actuator resources should be used in an equal manner by both interfaces. Both HMIs are visually perceived, and both trigger the same kind of manual responses. However, as the information processing framework of the SRK model suggests, cognitive resources should be used less when skill-based and, thus, automatized behavior is enabled and the human uses the given information as a signal instead of being forced to use it as a sign. Therefore, driving using the ecological interface should generate a cognitive load that is comparably low.

Because the HMIs used in this study influence the driving task and theoretically use more or less cognitive resources, it should be possible to measure these differences by the degree to which resources are allocated to the DRT. This idea is also explained in [Fig. 5](#), where the variables are kept constant and the only variation is between the conventional speedometer and ecological distance speedometer. A less demanding HMI should allow for more cognitive control to be allocated to the DRT.

Based on this DRT, the paper should validate that the ecological distance speedometer reduces cognitive load. Moreover, this ecological interface is also expected to improve the control over the vehicle by using an ecological display that elicits the appropriate behavior without conscious processing.

Accordingly, two research questions are addressed in this experiment. Does the ecological interface, following the previous results by [Schewe and Vollrath \(2019\)](#), enhance driving performance? Moreover, does the ecological interface reduce the cognitive load?

Speed control with a normal speedometer



Speed control with a distance speedometer

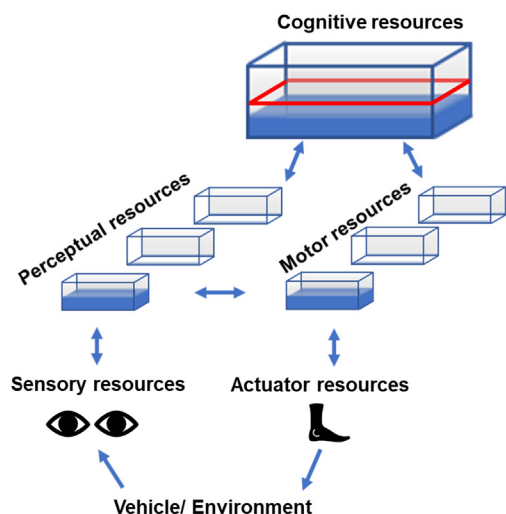


Fig. 4. Hypothetical use of cognitive resources, comparing a normal speedometer with an ecological distance speedometer with the help of a filling level marking. Adapted from [ISO Standard 17488 \(2016\)](#).

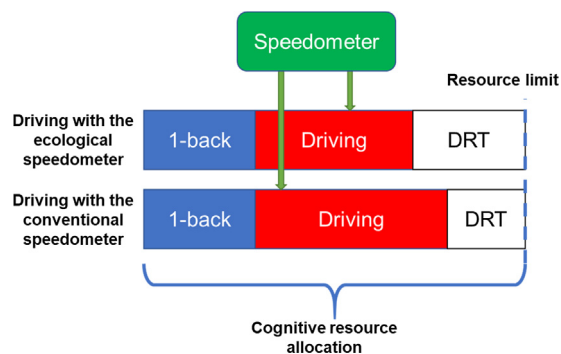


Fig. 5. Rationale of the testing approach of the detection response task (DRT) with hypothetical resource allocations for different human-machine interfaces.

2. Method

2.1. Design and procedure

This experimental study was conducted as a within-subject $2 \times 2 \times 2$ factorial design. The three independent variables are the HMI, ISA warnings, and the 1-back task. Each of these three factors has two levels. The two levels of HMI are driving with the conventional speedometer and driving with the ecological speedometer. The two levels of ISA warnings are that the ISA is either on or off. Lastly, the 1-back task was used to either add workload or not. A graphic representation of the study design is illustrated in Fig. 6.

Every participant drove in all eight conditions of the $2 \times 2 \times 2$ design. These were presented in four rides (all combinations of HMI and ISA warnings), which were done with the 1-back task either in the first half (Blocks 1 and 2) or second half (Blocks 3 and 4) of the track. Therefore, in order to be able to interpret the possible effects, the track consisted of two identical halves. Moreover, to avoid the effects of order and first-order carry-over effects, we used the matrix from MacFie, Bratchell, Greenhoff, and Vallis (1989). Based on this matrix, one participant drove in all conditions, each condition occurred equally frequently at each position of the experiment, and each condition was preceded equally frequently by each of the other conditions.

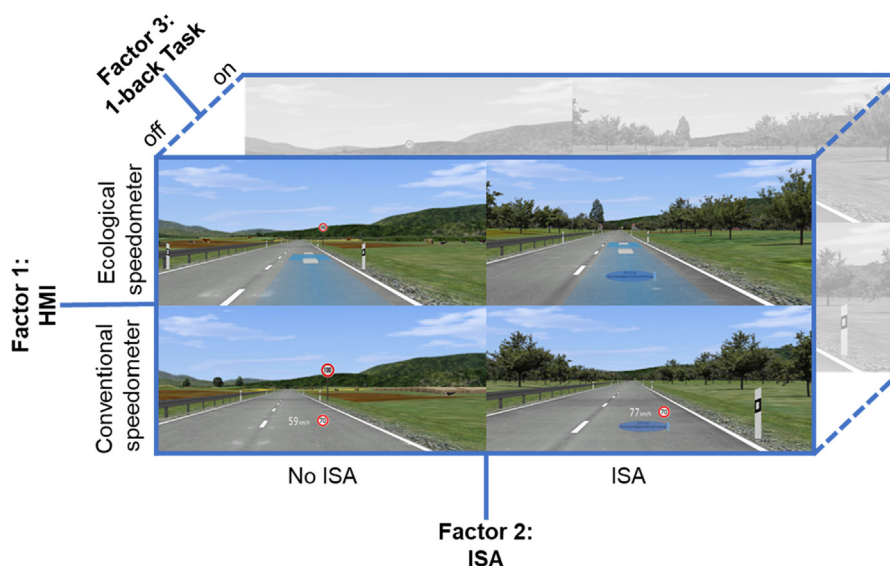


Fig. 6. The $2 \times 2 \times 2$ design of the study.

2.2. Participants

The final sample consisted of 49 participants. Six of the initial 55 participants had to be excluded due to failure of the detection response task or due to simulator sickness. The participants were recruited from an internal university database and by using an e-mail distribution list. Table 1 presents details on the number of participants, their sex, their yearly mileage, and their number of years licensed.

Participants had normal vision or else used corrective lenses. The 28 females had a mean age of 30 years ($SD = 11$), while the 21 males had a mean age of 38 years ($SD = 19$). The females drove on average 8 thousand kilometers per year ($SD = 8$), while the males drove on average 14 thousand kilometers per year ($SD = 10$). The research complied with the German Psychological Societies' adaptation of the American Psychological Association's Code of Ethics and was approved by the Institutional Review Board of the TU Braunschweig. Informed consent was obtained from each participant.

2.3. Simulator

The fixed-base half-vehicle mock-up simulator of the Department of Engineering and Traffic Psychology at the TU Braunschweig consists of a mock-up cockpit and an array of three screens (2×2 m), in addition to corresponding LCD projectors (1920×1080 pixels each) that cover a 180° field of vision from the driver's seat, which is approximately 2 m away from the center of the front screen. Two identical 7" screens were used as side mirrors, but no rear-view mirror was included. The head-up display was simulated on the front screen. The steering wheel, gas pedal, and brake pedal were equipped with force feedback. An automatic gearbox was simulated, and driving sounds were played by a surround-sound system. SILAB 5.0 (WIVW GmbH, <https://wivw.de/en/silab>) was used as the simulator software. Fig. 7 shows the configuration in order to get a feeling for the fidelity of the simulator.

Table 1
Demographics of the sample.

Sex	n	Yearly mileage (km in thousands)		Age		Years licensed	
		M	SD	M	SD	M	SD
Female	28	8	8	30	11	12	11
Male	21	14	10	38	19	21	19

Note. M = mean, SD = standard deviation.

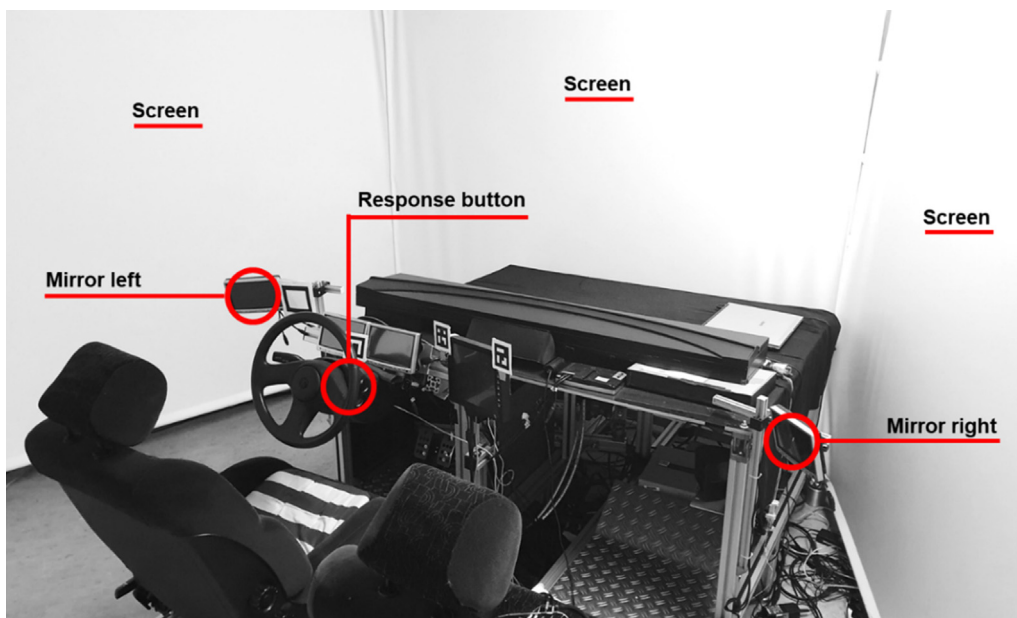


Fig. 7. Simulator setup.

2.4. Scenario

The scenario was used to provide a steady environment in which the factors of our experiment could be tested. It consisted of a rural road with speed limits of 50, 70, and 100 km/h. The speed changes only occurred on straight sections of the track. However, curves and varying scenery were also simulated, with matching speed limits. One ride took about 12 min and included a preceding familiarization run. The familiarization runs were implemented to train the 1-back task as well as getting used to the HMIs. The traffic density was set to low. Fig. 8 depicts the track, with its speed limits and curves. Because the effects of specific speed changes were not of interest for this study, the possible change from 50 to 100 km/h occurred just once between Blocks 2 and 3. However, all other combinations were implemented for the first and second half of the track. Both halves were identical to allow for the comparison of effects when combined with the 1-back task.

2.5. Dependent variables

2.5.1. Human-machine interface

Both the regular speedometer and ecological interface were presented in the head-up display. To provide comparable information and thus help the articulatory control process (Baddeley, Lewis, & Vallar, 1984), the current speed limit was visualized using both HMIs (Fig. 9).

2.5.2. Intelligent speed assistance warnings

In addition, an ISA was either present or not. The ISA system simply warned the driver both acoustically (with a “beep”) and visually (Fig. 9), if the driven speed was more than 5 km/h above the limit.

2.5.3. 1-Back task

A 1-back task was used to systematically induce cognitive workload. This audio task required the participants to listen to numbers, remember them, and repeat them (Mehler et al., 2009). The difficulty level for our study was set to moderate and involved repeating the number stated one step previously (1-back). We decided to use this moderate task because we wanted to ensure a low error rate to be able to compare the effects generated by the task. Moreover, participants were instructed to prioritize the 1-back task over other tasks. The 1-back task required the encoding of information, storage, rehearsal, temporal ordering, inhibition, and finally, response. The participants were required to respond verbally, and their responses were recorded manually by the experimenter. The participants were instructed to perform this task accurately and precisely. Missed and incorrect responses were coded as errors and were used to control the amount of induced workload.

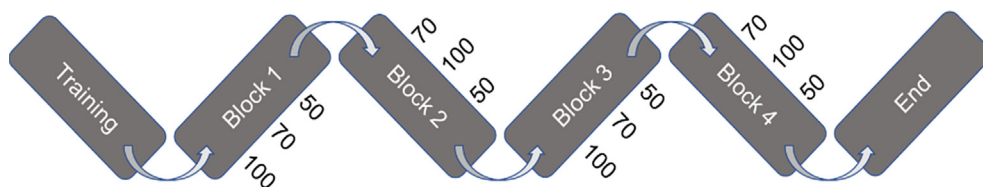


Fig. 8. The track with speed limits in km/h.

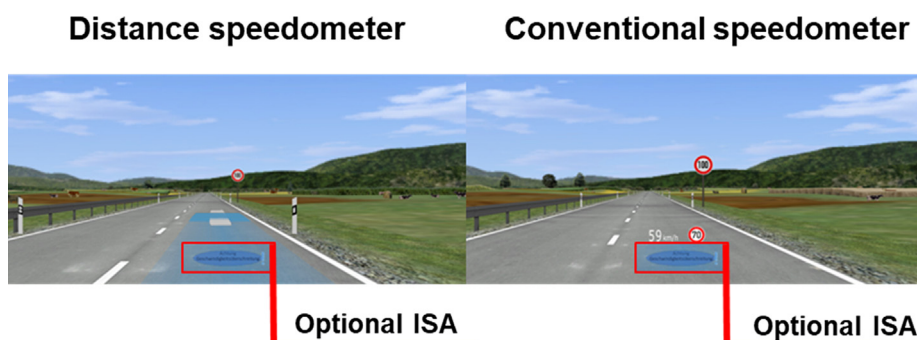


Fig. 9. Human-machine interface and intelligent speed assistance (ISA) design.

2.6. Operationalization of the dependent variables

2.6.1. Driving performance

The lateral driving performance was assessed by the standard deviation of the lateral position (SDLP), which represents “weaving.” This measure, introduced by O’Hanlon, Haak, Blaauw, and Riemersma (1982), is known to be a valid measure of driving performance with high test–retest reliability. To assess longitudinal driving performance, the standard deviation of speed deviation from the speed limit was also analyzed.

2.6.2. Cognitive workload

The original DRT is described in ISO 17488 (2016) and is commonly used to measure the attentive effects of cognitive workload induced by a secondary task (e.g., Krause, Conti, Späth, & Bengler, 2014). The DRT used in this experiment provided a vibration next to the participants’ clavicles. It was implemented by a mobile application (<https://github.com/InstituteOfErgonomics/MDT>) on a Samsung Galaxy S3 smartphone and caused the device’s hardware to vibrate. A button was attached to the steering wheel and was connected to the smartphone. The participants were instructed to respond as quickly and accurately as possible to every vibration they detected. The interstimulus interval varied randomly from 3 to 5 s. The response signals remained on for 1000 ms or until the button was pressed. The reaction times and hits were used as relevant measures for cognitive load. According to ISO, the reaction times particularly reflect the cognitive workload. The mobile application was equipped with a long press alarm, prohibiting random responses. Fig. 10 shows the experimental setup.

2.7. Statistical analysis

We used a three-way repeated-measure analysis of variance (ANOVA; $2 \times 2 \times 2$) to separately identify the effects of the independent variables on lateral and longitudinal vehicle control as well as the effects on the DRT reaction times and hit rates. Due to good readability and comprehensibility, we solely report findings on the untransformed raw data. However, because all subsets of the data had slightly skewed distributions (Shapiro-Wilk tests; $p > .05$) and contained few data points that were identified as outliers, we also applied a logarithmic transformation and repeated the analysis on this normalized data. As the significant results and effect sizes did not diverge meaningfully, the reported findings on the raw data are considered valid. The data were preprocessed using R Statistics (v. 3.4.3, R Core Team, <https://www.r-project.org/>) and were analyzed using SPSS statistics (v. 26, IBM Corp, <https://ibm.com/spss/>).

2.8. Hypotheses

According to our research questions, the hypotheses were formulated as follows:

- Hypothesis A: There is a main effect for the HMI on SDLP.
- Hypothesis B: There is a main effect for the HMI on the standard deviation of speed deviation.
- Hypothesis C: There is a main effect for the HMI on DRT reaction times.
- Hypothesis D: There is a main effect for the HMI on DRT hit rates.

Additionally, the main effects of the ISA and 1-back task and the interaction effects with these factors were analyzed in an exploratory way to discuss the practical relevance of the ecological interface.

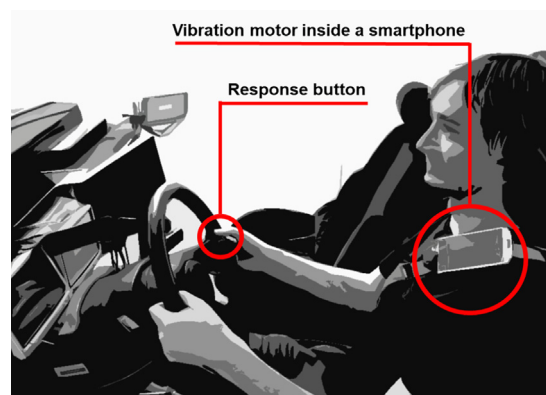


Fig. 10. Configuration of the detection response task.

3. Results

The results are structured into two parts. First, the findings on driving performance are presented because they could have influenced the way the attentional effects of cognitive load must be interpreted. Second, the assessment of the cognitive workload itself is presented. To simplify the figures, the conventional speedometer is named conventional and the ecological distance speedometer is named ecological. In [Appendix B](#) the F -values, p -values, and effect sizes for all possible main and interaction effects are presented. Additionally, at the beginning of the results, the analysis of the 1-back task performance and number of ISA warnings is presented.

3.1. Number of 1-back task failures

To interpret the overall results properly, we checked for 1-back failures, which occurred on average 3.7 times ($SD = 4.8$) out of the 70 trials per condition under which the 1-back task actually was given. To statistically test whether the HMI or presence of ISA influenced the number of 1-back failures, a 2×2 repeated-measure ANOVA was conducted. The conditions under which no 1-back or ISA was given were not part of this specific analysis. Because no significant main effects or interactions were found for the 1-back failures (all $p > .56$, all $\eta^2 < 0.01$), we are confident that the 1-back task served as a stable source of workload and thus does not influence the interpretability of the other results. Results of the 1-back failures are also summarized in [Appendix A](#).

3.2. Number of intelligent speed assistance interventions

Second, ISA interventions were counted and analyzed. To statistically test whether the HMI or 1-back task influenced the number of ISA warnings, another 2×2 repeated-measure ANOVA was conducted. As for the 1-back task failures, the conditions under which no ISA was given were not part of this specific analysis. The ANOVA on ISA interventions shows a main effect of HMI ($F [1, 48] = 224.23$, $p < .01$, $\eta^2 = 0.82$). As [Fig. 11](#) shows, the system had to intervene more often with the conventional interface than with the ecological interface. No other main or interaction effects were found (all $p > .31$, all $\eta^2 < 0.02$). All results are also summarized in [Appendix A](#).

3.3. Driving performance: Standard deviation of the lateral position

The ANOVA on the SDLP shows a main effect of HMI ($F [1, 48] = 31.34$, $p < .01$, $\eta^2 = 0.40$). No significant interactions or other main effects were found (all $p > .17$, all $\eta^2 < 0.04$). As [Fig. 12](#) shows, driving with the ecological interface leads to a reduced SDLP in all conditions; thus, Hypothesis A is supported. The SDLP is neither influenced by the 1-back task nor the ISA warnings.

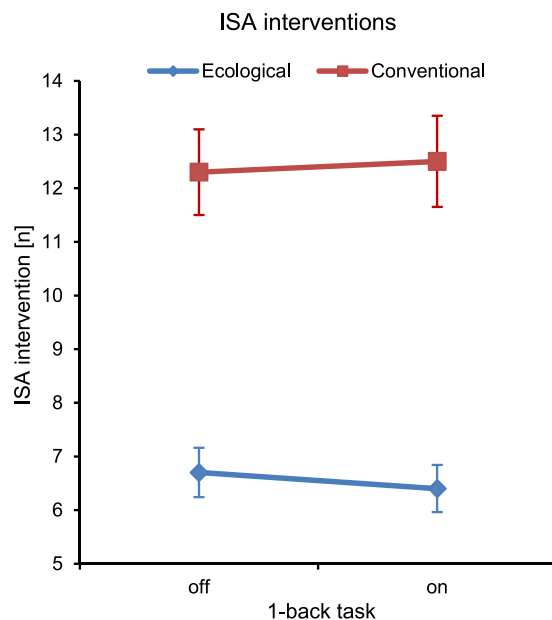


Fig. 11. Number of intelligent speed assistance (ISA) interventions (mean and standard error).

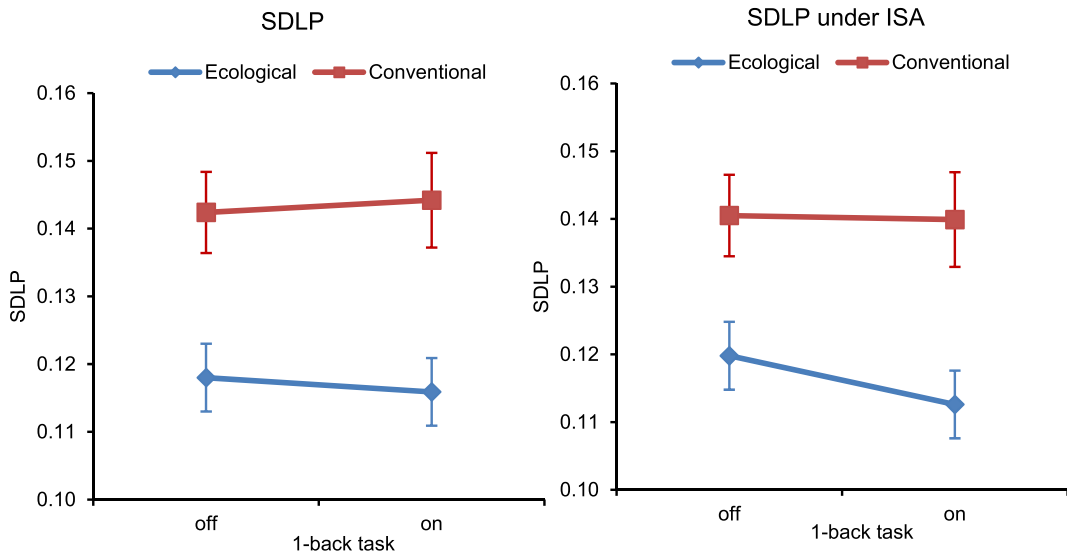


Fig. 12. SDLP (mean and standard error).

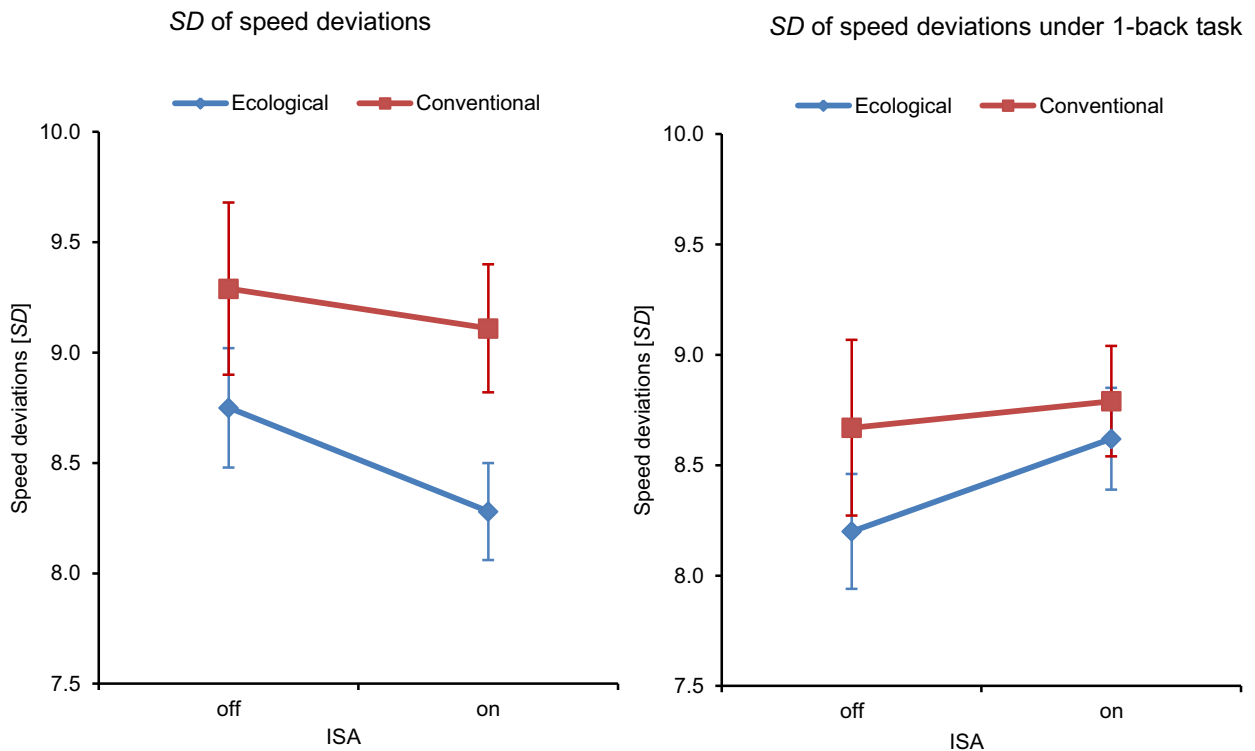


Fig. 13. Standard deviation (SD) of speed deviations (mean and standard error).

3.4. Driving performance: Speed deviations

The ANOVA on speed deviations demonstrates a main effect of HMI ($F[1, 48] = 5.51, p = .02, \eta^2 = 0.10$). A small tendency of a two-way interaction also exists between the 1-back task and ISA, ($F[1, 48] = 3.08, p = .09, \eta^2 = 0.06$). Computing an ANOVA based on the normalized data, this tendency becomes significant. However, no other significant interactions or main effects were found (all $p > .19$, all $\eta^2 < 0.04$). As Fig. 13 shows, the ecological interface reduces the variability of speed deviations. Thus, Hypothesis B is supported by the experimental results. Regarding the tendency of interaction, it seems that without the

1-back task, the ISA reduces the standard deviation of the speed deviations. With the additional workload of the 1-back task, the effect is more likely to be the opposite.

3.5. Detection response task: Reaction times

The three-way repeated-measure ANOVA on reaction times shows that there is a statistically significant main effect of the HMI ($F[1, 48] = 7.47, p < .01, \eta^2 = 0.14$). However, no significant interactions or other main effects were found (all $p > .32$, all $\eta^2 < 0.02$). Fig. 14 shows that the ecological interface reduces DRT reaction times, which indicates reduced cognitive workload. This result supports Hypothesis C. Because no significant main effect was found for the 1-back task, which should increase mental workload and could lead to an increase in DRT, the effect of the HMI appears substantial.

3.6. Detection response task: Hit rates

The ANOVA on hit rates indicates a tendency for a main effect of HMI ($F[1, 48] = 3.28, p = .08, \eta^2 = 0.06$). As Fig. 15 shows, hit rates are somewhat higher with the ecological interface. However, this tendency could not be replicated on the transformed dataset. Thus, Hypothesis D is not supported by our experiment. No significant interactions or other main effects were found (all $p > .43$, all $\eta^2 < 0.01$).

4. Discussion

This paper aimed to examine whether an ecological interface reduces cognitive load, even though it improves performance. Therefore, we used the example of HMIs for speed control and tested a conventional speedometer against an ecological distance speedometer. To assess a larger and more valid set of possible situations, we included an ISA system and a 1-back task in our study. The first research question examines whether driving performance is enhanced by using the ecological distance speedometer under diverse driving situations. The second research question addressed whether the ecological interface reduces cognitive load.

To control for confounding variables, we ensured a steady test environment. Additionally, we checked for consistent performance on the 1-back task, which provided a consistent cognitive load when applied. After checking the number of ISA interventions, a main effect of HMI was found. Using the ecological interface, significantly fewer ISA interventions occurred. This indicates, that unintentional speeding is more likely while using the conventional speedometer.

However, examining the ISA and 1-back task as factors, no significant main or interaction effects on the dependent variables were found. Only a slight tendency for a crossover interaction was found. The effect of the ISA on the variability of speed deviation is the opposite, depending on whether a 1-back task was performed. This slight tendency might be explained by the fact that the induced load might hinder the proper adaption to the ISA warnings. This also supports the argumentation

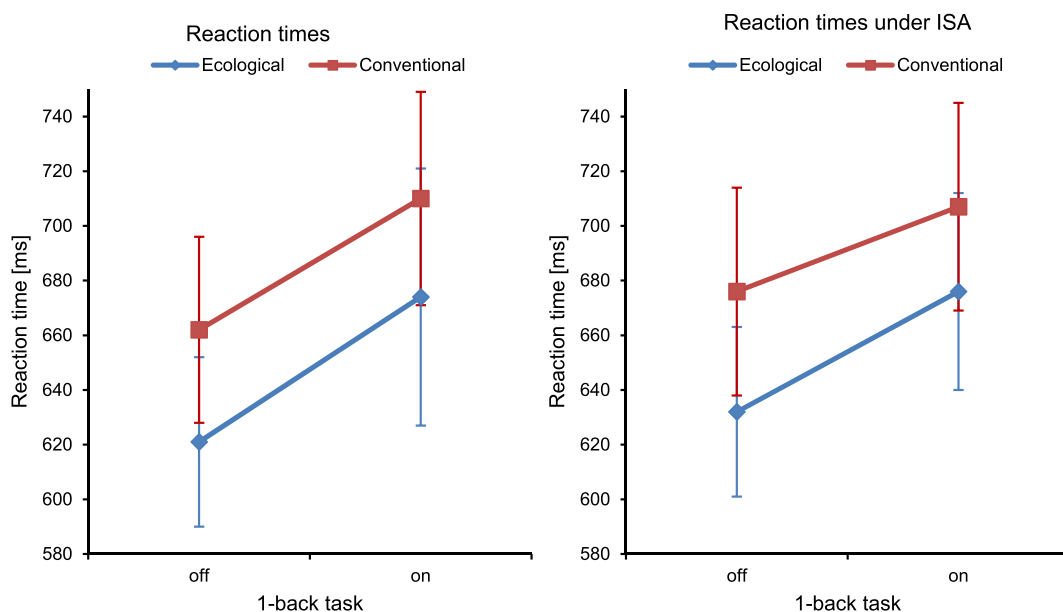


Fig. 14. Reaction times (mean and standard error).

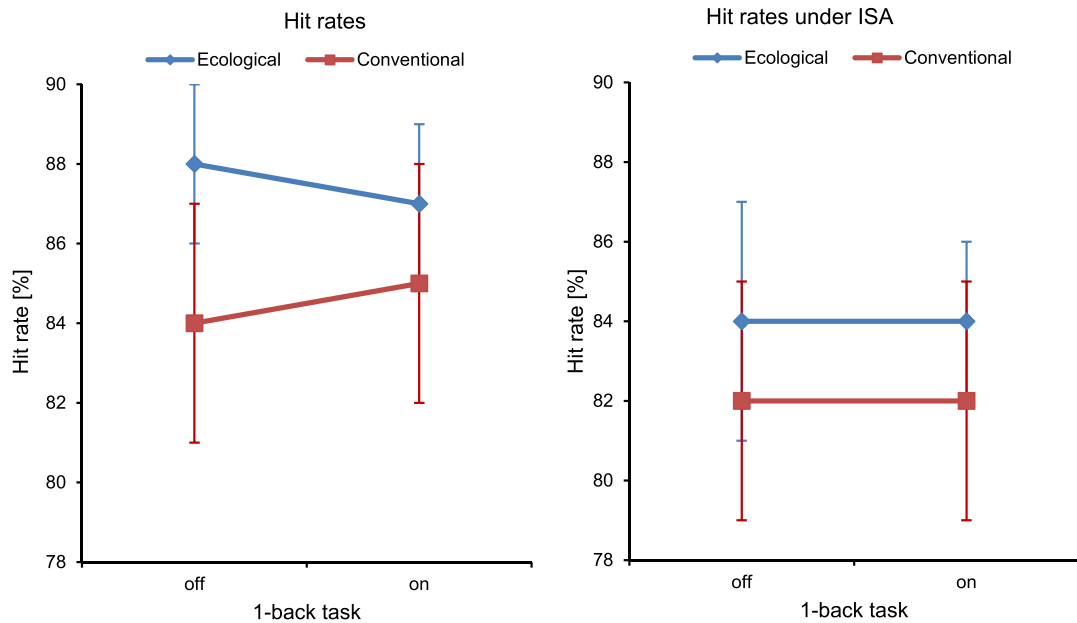


Fig. 15. Hit rates (mean and SE).

that warnings trigger rule-based, nonautomatized behavior, which must compete for a small share of cognitive resources. In contrast, automated, skill-based behavior appears to be unaffected by cognitive load.

The ecological interface has a positive effect on lateral control, namely, SDLP. The drivers using the ecological interface were better able to maintain the lane in comparison to the drivers using a conventional speedometer. The augmented carpet of the ecological interface may have provided better information on the vehicle's current position in the lane. For automatized lane maintenance, the interface provides a better perceivable consistent cue.

Moreover, a positive effect of the ecological interface was found for speed control both with and without ISA. Therefore, the ecological interface enables better speed control without the necessity of using other systems, such as an ISA. The finding that the ecological interface enables the drivers to follow the set speed more constantly, indicates that their behavior is more automatized. Beyond this indication, the following paragraph explains why it is highly likely that the ecological interface enables the skill-based control of speed.

The reactions in the DRT were significantly faster with the ecological interface. Descriptively, the hit rate was also somewhat larger with the ecological interface. It appears that the ecological interface requires less cognitive resources than a conventional speedometer and that these cognitive resources can then be used for the DRT. As described in the introduction of this paper, the interface might be used as a signal. Descriptively, the 1-back task lengthens the reaction times, but in comparison to the HMI factor, no main effect was found. This facilitates understanding how meaningful the differences are in terms of cognitive control demand caused by the distance speedometer. However, following the cognitive control hypothesis, cognitively loaded drivers tend to resort to their individual, automatized behavioral patterns than to directly decrease their objective performance. Therefore, the measured differences do not directly relate to a safety gain. In most situations, the automatized behavior works very well, until the driver relies on a routine that fails to be appropriate for a situation. Depending on the situation, speeding, missing right of way signs, or misinterpreting the trajectories of vulnerable road users might be fatal. Nevertheless, in terms of crash causation, other measures, such as glances off the road, are much more directly related to fatalities (e.g., Victor et al., 2015).

Taken together, the ecological interface reduces the use of cognitive resources while simultaneously enabling better control of the vehicle. It improves the lateral vehicle control because it provides visual support for the lane-maintaining behavior which is already automated. Additionally, it enables the driver to control the speed and, more importantly, the changes in speed automatically. Thus, the information presented by the ecological interface reduces the cognitive load in comparison to a conventional speedometer. This is in keeping with the reasoning that a conventional speedometer requires rule-based behavior, whereas the information on the ecological interface can act as a direct input signal for procedures, which is a characteristic of skill-based and, thus, automatized behavior.

All of the results of our study are interrelated. Faster reaction times in the DRT relate to skill-based behavior, and the enhanced speed control also relates to skill-based behavior. Under cognitively demanding circumstances, the ecological interface might be able to replace individual, automatized routines by automatized behavior that follows the objectively controlled signals of the HMI. This might be a substantial safety gain and, for example, prevent unintentional speeding. Moreover, because it generally demands less cognitive control, it might enhance the driver's ability to better compute the

signs and symbols of his or her environment. Even if no direct crash causation exists, this extra cognitive control might prevent the onset of an event cascade that could lead to an accident.

Regarding the methods used, it is necessary to validate the findings in real traffic. The head-up display in the driving simulator is directly integrated into the scenery and appears on the screen. The impression is almost the same as that of a real head-up display but is not completely the same. In addition, the environment within the driving simulator is also clearly different from a real-world environment. Therefore, field tests of this system are a necessary next step in examining the effects of such an ecological interface. Moreover, it would be interesting to examine the interface in more complex driving situations that are likely to create safety-critical situations. For example, with a 2-back or 3-back task the effects of the ecological interface may be stronger. To avoid endangering drivers, this should first be tested in another driving simulator study. Lastly, long-term effects of the ecological distance speedometer should be tested. However, in summary, strong evidence demonstrates that the ecological interface design applied to the speed control domain reduces cognitive load. This finding should be used for further research on the ecological interface design and its applied purposes.

5. Outlook

The reduction of cognitive load provided by ecological interfaces should be used in our increasingly complex world. With respect to traffic and engineering psychology, interface design for future vehicles should consider the use of ecological interfaces. With current developments in augmented reality, such as holograms et cetera, the possibilities of this kind of interface are rapidly expanding. Future traffic systems may dispense with a physical infrastructure, such as signs, and rather use a virtual infrastructure. As this paper demonstrates, ecological interface design effectively reduces cognitive load and thus increases safety margins to cope with other possible tasks. Moreover, drivers can be prevented from unconsciously using unsafe, automatized behavioral routines.

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CRedit authorship contribution statement

Frederik Schewe: Conceptualization, Methodology, Formal analysis, Investigation, Project administration, Visualization, Writing - original draft, Writing - review & editing. **Mark Vollrath:** Supervision, Writing - review & editing.

Declaration of Competing Interest

None.

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Appendix A

	<i>F</i>	<i>Hypothesis df</i>	<i>Error df</i>	<i>p</i>	η^2
<i>1-back failures</i>					
HMI	0.158	1.000	48.000	0.693	0.003
ISA	0.337	1.000	48.000	0.564	0.007
HMI*ISA	0.012	1.000	48.000	0.913	0.000
<i>ISA warnings</i>					
(*)HMI	224.230	1.000	48.000	0.000	0.824
1-back	.004b	1.000	48.000	0.949	0.000
HMI*1-back	1.048	1.000	48.000	0.311	0.021

(*) = significant.

Appendix B

	<i>F</i>	<i>Hypothesis</i> <i>df</i>	<i>Error df</i>	<i>p</i>	η^2
<i>SDLP</i>					
(*)HMI	31.337	1.000	48.000	0.000	0.395
1-back	0.601	1.000	48.000	0.442	0.012
ISA	0.928	1.000	48.000	0.340	0.019
HMI*1-back	1.911	1.000	48.000	0.173	0.038
HMI*ISA	0.309	1.000	48.000	0.581	0.006
1-Back*ISA	1.179	1.000	48.000	0.283	0.024
HMI/1-back*ISA	0.151	1.000	48.000	0.699	0.003
<i>Speed deviations</i>					
(*)HMI	5.509	1.000	48.000	0.023	0.103
1-back	0.071	1.000	48.000	0.792	0.001
ISA	1.778	1.000	48.000	0.189	0.036
HMI*1-back	0.009	1.000	48.000	0.924	0.000
HMI*ISA	1.524	1.000	48.000	0.223	0.031
1-Back*ISA	3.075	1.000	48.000	0.086	0.060
HMI/1-back*ISA	0.696	1.000	48.000	0.408	0.014
<i>Reaction times</i>					
(*)HMI	7.468	1.000	48.000	0.009	0.135
1-back	1.017	1.000	48.000	0.318	0.021
ISA	0.477	1.000	48.000	0.493	0.010
HMI*1-back	0.190	1.000	48.000	0.665	0.004
HMI*ISA	0.008	1.000	48.000	0.929	0.000
1-Back*ISA	0.270	1.000	48.000	0.606	0.006
HMI/1-back*ISA	0.058	1.000	48.000	0.810	0.001
<i>Hit rates</i>					
(*)HMI	3.284	1.000	48.000	0.076	0.064
1-back	0.027	1.000	48.000	0.871	0.001
ISA	0.643	1.000	48.000	0.427	0.013
HMI*1-back	0.119	1.000	48.000	0.732	0.002
HMI*ISA	0.518	1.000	48.000	0.475	0.011
1-Back*ISA	0.138	1.000	48.000	0.712	0.003
HMI/1-back*ISA	0.551	1.000	48.000	0.462	0.011

(*) = significant.

Appendix C. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trf.2020.05.009>.

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